

Education

New York University, New York, NY

Aug 2024 – May 2026

Master of Science in Computer Engineering (GPA: 3.9/4.0)

Coursework: Machine Learning, Deep Learning, Big Data, System Design, Computer Architecture, Data Structures

Experience

- CampusMesh** Jan 2026 – Present
AI Software Engineer
– Developed scalable **Node.js microservices** powering real-time academic workflows across study groups, messaging, calls, notifications, and profile management, backed by **MySQL (Sequelize)** and **DynamoDB**.
– Deployed **serverless architecture** on AWS using **Serverless Framework** with **Lambda, API Gateway, S3, and CloudFront** across **4 environments** (local, dev, staging, production); enforced security with **KMS-managed keys, SSM Parameter Store**, and **least-privilege IAM**.
– Worked on **LLM-based retrieval pipelines (RAG)** for CampusIQ (AI Assistant), leveraging **embedding-based semantic search** and structured data indexing to improve contextual accuracy by **30%**.
- Amazon** May 2025 – Aug 2025
Software Development Engineer Intern
– Deployed a **Retrieval-Augmented Generation (RAG)** pipeline with an **agentic workflow** that autonomously retrieves operational logs to reason over incident details, generating context-aware summaries and remediation steps.
– Implemented a **Model Context Protocol (MCP)** server for efficient tool-augmented retrieval, cutting AWS Athena query time and compute costs by **40%**.
– Built a high-throughput, event-driven incident triage system on **AWS ECS Fargate, SQS, SNS**, decoupling distributed message processing and reducing response latency by **50%**.
– Provisioned resilient **Infrastructure-as-Code** using **AWS CDK** on auto-scaling ECS clusters, ensuring **99.9%** availability under 2x peak load.
- New York University (Novel AI Technologies, Inc.)** Aug 2024 – Present
AI Software Engineer
– Architected an **NSF-funded (\$100K)** AI-powered health platform for **iOS** and **Android** ingesting real-time wearable telemetry; implemented **time-series anomaly detection** using **Isolation Forest** and **LSTM** models.
– Built end-to-end **computer vision pipelines (YOLOv8, EfficientNet)** for user-submitted food images, integrated with **GPT-4 vision workflows** for nutritional estimation and dietary feedback.
– Developed a full-stack insurance analytics product (**React, Node.js, REST APIs**) integrating **LLM-based document analysis (GPT-4 + LangChain)** for underwriting insights.
– Engineered **ETL pipelines** transforming **NoSQL** into **PostgreSQL** with concurrency control; processed **250K+** records and powered **AWS QuickSight** dashboards.
- PricewaterhouseCoopers (PwC)** Aug 2021 – Aug 2024
Senior Software Engineer
– Designed a distributed backend on **Microsoft Azure** integrating with **Microsoft Dynamics** and supporting **ML inference pipelines** for healthcare analytics.
– Built an **NLP-powered OCR pipeline** using **LayoutLM transformer** and Azure Form Recognizer to automate document data extraction, reducing processing time by **75%**.
– Optimized data ingestion using **parallel processing** on **100K+** records, reducing execution from **5 hours to 1 hour**.
– Led code reviews and mentored engineers on **software design and test-driven development**, reducing production bugs by **25%**.

Projects

- InviLite (AI-Driven Data Engineering & Analytics Platform)** | *AWS, SageMaker, Bedrock, FAISS, Python*
– Architected a cloud-native data pipeline (Kinesis, Lambda, S3 Bronze/Silver/Gold, Glue, DynamoDB) ingesting structured and unstructured enterprise data into governed ML pipelines.
– Built a **RAG system** (SVM classification + FAISS vector indexing + **Bedrock LLM**), cutting incident resolution from **1–2 hours to minutes** and lifting root-cause classification accuracy from **56% to 92%** via feature engineering and model retraining.
- RoBERTa Fine-Tuning with Custom LoRA** | *Python, PyTorch, Hugging Face, AGNews*
– Architected a **parameter-efficient fine-tuning (PEFT)** pipeline using a custom **Low-Rank Adaptation (LoRA)** implementation to optimize a **RoBERTa-base** model for news classification.
– Optimized model efficiency by freezing pretrained weights and injecting trainable rank-decomposition matrices, reducing trainable parameters to under **1M (934K)** while achieving **83.45%** accuracy on the private leaderboard.
- High-Efficiency ResNet Image Classifier** | *Python, PyTorch, Torchvision, Matplotlib*
– Designed and implemented a specialized **ResNet** architecture optimized for the **CIFAR-10** dataset, adhering to a strict **5M parameter** limit for edge-ready deployment.
– Enhanced model convergence by integrating the **Lookahead optimizer** and advanced data augmentation, with real-time visualization via **TensorBoard**.

Skills

- Programming Languages:** Python, Java, C++, JavaScript (ES6), TypeScript, SQL, Go
- AI/ML:** PyTorch, TensorFlow, LLM Fine-tuning, RAG (LangChain, FAISS), Transformers, NLP, Hugging Face, Computer Vision (YOLOv8, EfficientNet), Time-Series Modeling, MLflow, SVM, Random Forest, Pandas, NumPy
- Web & Backend:** React, Node.js, Express.js, REST API, GraphQL API, Microservices
- Databases:** MySQL, PostgreSQL, MongoDB, DynamoDB
- Cloud & DevOps:** AWS (Lambda, SageMaker, Bedrock, ECS, S3, CloudWatch, CDK, API Gateway, IAM), Azure (AI Services, Function Apps), Docker, Kubernetes, CI/CD, Terraform
- Tools & Platforms:** Git, Linux, Jupyter, Postman, Cursor, GitHub Copilot, Model Context Protocol (MCP)

Publications

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques (*IEEE*)

An Aquila-optimized SVM Classifier for Diabetes Prediction (*IEEE*)

Certifications

AWS Cloud Practitioner

Stanford Machine Learning (Coursera)

Azure Data Fundamentals